

Chapter Eight

Variation of Parameters

8.1 Method of Variation of Parameters

We consider a linear second order equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = b(x) \quad \dots (1)$$

Dividing both sides of (1) by the leading coefficient $a_2(x)$, we reduced it into standard form

$$y'' + P(x)y' + Q(x)y = f(x) \quad \dots (2)$$

where the coefficient functions $P(x) = \frac{a_1(x)}{a_2(x)}$, $Q(x) = \frac{a_0(x)}{a_2(x)}$ and $f(x) = \frac{b(x)}{a_2(x)}$

are continuous on some common interval $a \leq x \leq b$.

Let the complementary function of (2) be

$$y_c = c_1 y_1(x) + c_2 y_2(x) \quad \dots (3)$$

where the coefficients c_1 and c_2 are arbitrary constants.

Now replacing the constants coefficients c_1 and c_2 by the variables coefficients $u(x)$

and $v(x)$, we let the particular integral be

$$y_p = u(x)y_1(x) + v(x)y_2(x) \quad \dots (4)$$

Differentiating (4) w. r. to x , we get

$$y'_p = u'(x)y_1(x) + u(x)y'_1(x) + v'(x)y_2(x) + v(x)y'_2(x), \quad \dots (5)$$

where we use primes to denotes differentiations.

Now, for simplicity, we impose the condition

$$u'(x)y_1(x) + v'(x)y_2(x) = 0 \quad \dots (6)$$

Then, then from (5), we have

$$y'_p = u(x)y'_1(x) + v(x)y'_2(x)$$

Differentiating w. r. to x , we get

$$y''_p = u'(x)y'_1(x) + u(x)y''_1(x) + v'(x)y'_2(x) + v(x)y''_2(x)$$

Substituting the values of y_p , y'_p and y''_p instead of y , y' and y'' in (2), we get

$$\begin{aligned} \{u'(x)y'_1(x) + u(x)y''_1(x) + v'(x)y'_2(x) + v(x)y''_2(x)\} + \{u(x)y'_1(x) + v(x)y'_2(x)\}P(x) \\ + \{u(x)y_1(x) + v(x)y_2(x)\}Q(x) = f(x) \end{aligned}$$

$$\begin{aligned} \Rightarrow u(x)\{y''_1 + P(x)y'_1 + Q(x)y_1\} + v(x)\{y''_2 + P(x)y'_2 + Q(x)y_2\} \\ + \{u'(x)y'_1(x) + v'(x)y'_2(x)\} = f(x) \end{aligned}$$

$$\Rightarrow u'(x)y'_1(x) + v'(x)y'_2(x) = f(x) \quad \dots (7)$$

Here, $y_1'' + P(x)y_1' + Q(x)y_1 = 0$ and $y_2'' + P(x)y_2' + Q(x)y_2 = 0$; since, $y_1(x)$ and $y_2(x)$ are two linearly independent solutions of the homogeneous differential equation (2), i.e., $y'' + P(x)y' + Q(x)y = 0$.

Now, solving equations (6) and (7), using **Cramer's rule**, for $u'(x)$ and $v'(x)$, we get

$$u'(x) = \frac{\begin{vmatrix} 0 & y_2(x) \\ f(x) & y_2'(x) \end{vmatrix}}{\begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}} \quad \text{and} \quad v'(x) = \frac{\begin{vmatrix} y_1(x) & 0 \\ y_1'(x) & f(x) \end{vmatrix}}{\begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}}$$

$$\Rightarrow u'(x) = -\frac{f(x)y_2(x)}{W[y_1(x), y_2(x)]} \quad \text{and} \quad v'(x) = \frac{f(x)y_1(x)}{W[y_1(x), y_2(x)]},$$

where $W[y_1(x), y_2(x)]$ denotes the wronskian of $y_1(x)$ and $y_2(x)$.

$$\therefore u(x) = -\int \frac{f(x)y_2(x)}{W[y_1(x), y_2(x)]} dx \quad \text{and} \quad v(x) = \int \frac{f(x)y_1(x)}{W[y_1(x), y_2(x)]} dx \quad \dots (8)$$

Therefore, the particular integral is

$$y_p = u(x)y_1(x) + v(x)y_2(x) \quad \dots (9)$$

where, $u(x)$ and $v(x)$ have been obtained from (8).

Thus, the general solution of (2) is

$$y = y_c + y_p$$

$$\text{or, } y = c_1y_1(x) + c_2y_2(x) + u(x)y_1(x) + v(x)y_2(x);$$

The method is used to find the general solution of differential equation by changing the constant coefficients of y_c into variable coefficients (i.e., functions of x) is known as variation of parameters.

Note: This method is applicable for all the non-homogeneous equations but specially when, the right member of the equation is not itself a particular solution of any homogeneous linear differential equation with constant coefficients.

Solved Problems:

Example-1 Solve the following equation by the method of variation of parameters: $\frac{d^2y}{dx^2} + y = \sec x$.

Solution Given equation $\frac{d^2y}{dx^2} + y = \sec x \quad \dots (1)$

The corresponding homogeneous equation is

$$\frac{d^2y}{dx^2} + y = 0 \quad \dots (2)$$

Let $y = e^{mx}$ be a trial solution of (2) then, the auxiliary equation of (2) is

$$m^2 + 1 = 0 \quad [\text{since } e^{mx} \neq 0]$$

$$\therefore m = \pm i$$

∴ The complementary function is $y_c = c_1 \cos x + c_2 \sin x$

Let the particular solution of (1) be

$$y_p = u(x)\cos x + v(x)\sin x \quad \dots (3)$$

Differentiating (3) w. r. to x , we get

$$y'_p = -u\sin x + u'\cos x + v\cos x + v'\sin x \quad \dots (4)$$

Now we impose the condition,

$$u'\cos x + v'\sin x = 0 \quad \dots (5)$$

The equation (4) becomes,

$$y'_p = -u\sin x + v\cos x \quad \dots (6)$$

Differentiating (6) w. r. to x , we get

$$y''_p = -u\cos x - u'\sin x - v\sin x + v'\cos x \quad \dots (7)$$

Now putting the values of y_p and y''_p in (1), we get

$$\begin{aligned} -u\cos x - u'\sin x - v\sin x + v'\cos x + u\cos x + v\sin x &= \sec x \\ \Rightarrow -u'\sin x + v'\cos x &= \sec x \quad \dots (8) \end{aligned}$$

Using Cramer's rule solving (5) & (8), we get

$u' = \frac{\begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}}$ $= \frac{0 - \sin x \cdot \sec x}{\cos^2 x + \sin^2 x}$ $\Rightarrow \frac{du}{dx} = -\tan x$ $\Rightarrow \int du = -\int \tan x dx$ $\therefore u(x) = \ln \cos x + c_3,$ where c_3 is arbitrary constant	$v' = \frac{\begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}}$ $= \frac{\cos x \cdot \sec x - 0}{\cos^2 x + \sin^2 x}$ $\Rightarrow \frac{dv}{dx} = 1$ $\Rightarrow \int dv = \int dx$ $\therefore v(x) = x + c_4,$ where c_4 is arbitrary constant
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Now putting the values of $u(x)$ and $v(x)$ in (3), we get

$$\begin{aligned} y_p &= (\ln \cos x + c_3)\cos x + [x + c_4]\sin x \\ \Rightarrow y_p &= c_3 \cos x + c_4 \sin x + \cos x \ln \cos x + x \sin x \end{aligned}$$

∴ The general solution is $y = y_c + y_p$

$$\begin{aligned} &= c_1 \cos x + c_2 \sin x + c_3 \cos x + c_4 \sin x + \cos x \ln \cos x + x \sin x \\ &= C_1 \cos x + C_2 \sin x + \cos x \ln \cos x + x \sin x \quad (\text{Ans.}) \end{aligned}$$

where, $C_1 = c_1 + c_3$ and $C_2 = c_2 + c_4$.

Example-2 Solve the following equations by the method of variation of Parameters:

(i) $\frac{d^2y}{dx^2} + 4y = 4\tan 2x$ (ii) $\frac{d^2y}{dx^2} + y = \tan x$

Solution

(i) Given equation $\frac{d^2y}{dx^2} + 4y = 4\tan 2x$... (1) The corresponding homogeneous equation $\frac{d^2y}{dx^2} + 4y = 0 \quad \dots (2)$ Let $y = e^{mx}$ be a trial solution of (2) then, the auxiliary equation is $m^2 + 4 = 0$ [since $e^{mx} \neq 0$] $\Rightarrow m = \pm 2i$ $\therefore$ The complementary function is $y_c = c_1 \cos 2x + c_2 \sin 2x.$ Let us seek a particular solution by variation of parameters.	We consider, $y_p = A \cos 2x + B \sin 2x \quad \dots (3)$ where both A and B are functions of x. Differentiating (3) w. r. to x, we get $y'_p = -2A \sin 2x + A' \cos 2x + 2B \cos 2x + B' \sin 2x \quad \dots (4)$ For the sake of simplicity, we get $A' \cos 2x + B' \sin 2x = 0 \quad \dots (5)$ $\therefore$ Equation (4) becomes, $y'_p = -2A \sin 2x + 2B \cos 2x \quad \dots (6)$ Again differentiating this w. r. to x, we get $y''_p = -4A \cos 2x - 2A' \sin 2x - 4B \sin 2x + 2B' \cos 2x \quad \dots (7)$
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Now putting the values of y_p, y''_p in (1), we get

$$\begin{aligned} -4A \cos 2x - 2A' \sin 2x - 4B \sin 2x + 2B' \cos 2x + 4(A \cos 2x + B \sin 2x) &= 4 \tan 2x \\ \Rightarrow -2A' \sin 2x + 2B' \cos 2x &= 4 \tan 2x \\ \Rightarrow -A' \sin 2x + B' \cos 2x &= 2 \tan 2x \quad \dots (8) \end{aligned}$$

Using Cramer's rule solving (5) & (8), we get

$$B' = \frac{\begin{vmatrix} \cos 2x & 0 \\ -\sin 2x & 2 \tan 2x \end{vmatrix}}{\begin{vmatrix} \cos 2x & \sin 2x \\ -\sin 2x & \cos 2x \end{vmatrix}} = \frac{2 \cos 2x \cdot \tan 2x - 0}{\cos^2 2x + \sin^2 2x} = 2 \sin 2x$$

$$\Rightarrow \int dB = \int 2 \sin 2x \, dx$$

$$\therefore B = -\cos 2x + c_4, \text{ where } c_4 \text{ arbitrary constant}$$

$$A' = \frac{\begin{vmatrix} 0 & \sin 2x \\ 2 \tan 2x & \cos 2x \end{vmatrix}}{\begin{vmatrix} \cos 2x & \sin 2x \\ -\sin 2x & \cos 2x \end{vmatrix}} = \frac{0 - 2 \sin 2x \cdot \tan 2x}{\cos^2 2x + \sin^2 2x}$$

$$= -2 \sin 2x \cdot \frac{\sin 2x}{\cos 2x}$$

$$\Rightarrow \frac{dA}{dx} = -2 \left(\frac{1 - \cos^2 2x}{\cos 2x} \right) = -2 \sec 2x + 2 \cos 2x$$

$$\Rightarrow \int dA = -\int 2 \sec 2x \, dx + \int 2 \cos 2x \, dx$$

$$\therefore A = -\ln \tan \left(\frac{\pi}{4} + x \right) + \sin 2x + c_3,$$

where c_3 is an arbitrary constant

Now putting the values of A and B in (3), we get

$$y_p = \left[-\ln \tan \left(\frac{\pi}{4} + x \right) + \sin 2x + c_3 \right] \cos 2x + [-\cos 2x + c_4] \sin 2x$$

$$\therefore y_p = c_3 \cos 2x + c_4 \sin 2x - \cos 2x \ln \tan \left(\frac{\pi}{4} + x \right)$$

$\therefore$ The general solution is $y = y_c + y_p$

$$= c_1 \cos 2x + c_2 \sin 2x + c_3 \cos 2x + c_4 \sin 2x - \cos 2x \ln \tan \left(\frac{\pi}{4} + x \right)$$

$$= C_1 \cos 2x + C_2 \sin 2x - \cos 2x \ln \tan \left(\frac{\pi}{4} + x \right),$$

where, $C_1 = c_1 + c_3$ & $C_2 = c_2 + c_4$. (Ans.)

(ii) [Try yourself]

Example-3 By the method of variation of parameters to solve: $(D^2 + 1)y = \sec x \tan x$.

Solution Given equation $(D^2 + 1)y = \sec x \tan x \quad \dots (1)$ The corresponding homogeneous equation $(D^2 + 1)y = 0 \quad \dots (2)$ Let $y = e^{mx}$ be a trial solution of (2) then, the auxiliary equation is $m^2 + 1 = 0$ [since $e^{mx} \neq 0$] $\therefore m = \pm i$ $\therefore y_c = c_1 \cos x + c_2 \sin x$ where c_1, c_2 are arbitrary constants.	Let the particular solution of (1) be $y_p = A \cos x + B \sin x \quad \dots (3)$ where both A and B are functions of x. Differentiating (3) w. r. to x, we get $y'_p = -A \sin x + A' \cos x + B \cos x + B' \sin x \quad \dots (4)$ Now we impose the condition, $A' \cos x + B' \sin x = 0 \quad \dots (5)$ $\therefore$ The equation (4) becomes, $y'_p = -A \sin x + B \cos x \quad \dots (6)$
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Differentiating (6) w. r. to x , we get $y''_p = -A \cos x - A' \sin x - B \sin x + B' \cos x$

Now putting the values of y_p, y'_p in (1), we get

$$-A \cos x - A' \sin x - B \sin x + B' \cos x + A \cos x + B \sin x = \sec x \tan x$$

$$\Rightarrow -A' \sin x + B' \cos x = \sec x \tan x \quad \dots (7)$$

Using Cramer's rule solving (5) and (7), we get

$$A' = \frac{\begin{vmatrix} 0 & \sin x \\ \sec x \tan x & \cos x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}} = \frac{0 - \sin x \cdot \sec x \cdot \tan x}{\cos^2 x + \sin^2 x}$$

$$\Rightarrow A' = -\tan x \tan x = -\tan^2 x$$

$$\Rightarrow \frac{dA}{dx} = -(\sec^2 x - 1) = -\sec^2 x + 1$$

Integrating this w. r. to x , we get

$$A = -\tan x + x + c_3$$

$$B' = \frac{\begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \tan x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}} = \frac{\cos x \cdot \sec x \cdot \tan x - 0}{\cos^2 x + \sin^2 x}$$

$$\Rightarrow \frac{dB}{dx} = \tan x$$

Integrating this w. r. to x , we get

$$B = -\ln(\cos x) + c_4,$$

where c_4 is an arbitrary constant

Now putting the values of A and B in (3), we get

$$y_p = (x - \tan x + c_3) \cos x + [c_4 - \ln(\cos x)] \sin x$$

$$y_p = c_3 \cos x + c_4 \sin x - \sin x + x \cos x - \sin x \ln(\cos x)$$

$\therefore$ The general solution is

$$y = y_c + y_p$$

$$= c_1 \cos x + c_2 \sin x + c_3 \cos x + c_4 \sin x$$

$$- \sin x + x \cos x - \sin x \ln(\cos x)$$

$$= C_1 \cos x + C_2 \sin x + x \cos x - \ln(\cos x) \quad (\text{Ans.})$$

where $C_1 = c_1 + c_3$ and $C_2 = c_2 + c_4 - 1$.

Example-4 Solve the following equation by the method of variation of parameters.

$$\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x$$

Solution The general solution is $y = y_c + y_p = C_1 \cos x + C_2 \sin x - x \cos x + \sin x \cdot \ln(\sin x)$ (Ans.)

Example-5 By the method of variation of parameters to solve

$$(D^2 - 2D + 1)y = e^x \ln x$$

Solution The general solution is

$$y = y_c + y_p = C_1 e^x + C_2 e^x + \frac{x^2 e^x}{2} \left(\ln x - \frac{3}{2} \right) \text{ (Ans.)}$$

Example-6 Apply the method of variation of parameters to solve

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 3y = \frac{1}{1 + e^{-x}}$$

Solution Given equation

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 3y = \frac{1}{1 + e^{-x}} \quad \dots (1)$$

The corresponding homogeneous equation is

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 3y = 0 \quad \dots (2)$$

Let $y = e^{mx}$ be trial solution of (2), then the auxiliary equation is

$$m^2 - 4m + 3 = 0 \quad [\text{since } e^{mx} \neq 0]$$

$$\Rightarrow m = 1, 3$$

$\therefore$ The CF is $y_c = C_1 e^x + C_2 e^{3x}$

Let the PI of (1) be $y_p = ue^x + ve^{3x} \quad \dots (3)$

where both u and v are functions of x .

Differentiating (3) w. r. to x , we get

$$y'_p = ue^x + u'e^x + 3ve^{3x} + v'e^{3x} \quad \dots (4)$$

Now we impose the condition,

$$u'e^x + v'e^{3x} = 0$$

$$\Rightarrow u' = -v'e^{2x} \quad \dots (5)$$

From equation (4), we have $y'_p = ue^x + 3ve^{3x}$

Differentiating (5) w. r. to x , we get

$$y''_p = ue^x + u'e^x + 9ve^{3x} + 3v'e^{3x}$$

Now putting the values of y_p, y'_p and y''_p in (1), we obtain

$$(ue^x + u'e^x + 9ve^{3x} + 3v'e^{3x}) - 4(ue^x + 3ve^{3x}) + 3(ue^x + ve^{3x}) = \frac{1}{1 + e^{-x}}$$

$$\Rightarrow u'e^x + 3v'e^{3x} = \frac{1}{1 + e^{-x}}$$

$$\Rightarrow -v'e^{2x}e^x + 3v'e^{3x} = \frac{1}{1 + e^{-x}} \quad [\text{using (5)}]$$

$$\Rightarrow 2v'e^{3x} = \frac{1}{1 + e^{-x}}$$

$$\Rightarrow v' = \frac{1}{2e^{3x}(1 + e^{-x})}$$

$$\Rightarrow v' = \frac{1}{2e^{2x}(1 + e^x)} \quad \dots (6)$$

$$\Rightarrow v' = \frac{1}{2z^2(z+1)} \quad [\text{We put } e^x = z] \quad \dots (7)$$

$$\text{Now let } \frac{1}{z^2(z+1)} = \frac{1}{z^2} + \frac{A}{z} + \frac{1}{z+1}$$

$$\Rightarrow 1 = (z+1) + Az(z+1) + z^2$$

Equating coefficient of z^2 from both sides, we get

$$0 = A + 1 \Rightarrow A = -1$$

Thus the equation (7) becomes,

$$\Rightarrow v' = \frac{1}{2} \left[\frac{1}{z^2} - \frac{1}{z} + \frac{1}{z+1} \right]$$

$$\Rightarrow \frac{dv}{dx} = \frac{1}{2} \left[\frac{1}{e^{2x}} - \frac{1}{e^x} + \frac{1}{e^x + 1} \right]$$

$$\Rightarrow \int dv = \frac{1}{2} \left[\int e^{-2x} dx - \int e^{-x} dx + \int \frac{e^{-x}}{1+e^{-x}} dx \right]$$

$$\therefore v = -\frac{1}{4}e^{-2x} + \frac{1}{2}e^{-x} - \frac{1}{2}\ln(1+e^{-x}) + C_4,$$

where C_4 is an arbitrary constant

Putting the value of v' from (6) into the equation (5), we get

$$u' = -\frac{1}{2(1+e^x)}$$

$$\Rightarrow \frac{du}{dx} = -\frac{e^{-x}}{2(1+e^{-x})}$$

$$\Rightarrow \int du = -\frac{1}{2} \int \frac{e^{-x}}{1+e^{-x}} dx$$

$$u = \frac{1}{2} \ln(1+e^{-x}) + C_3$$

Now putting the values of u and v in (3), we get

$$y_p = \left[\frac{1}{2} \ln(1+e^{-x}) + C_3 \right] e^x$$

$$+ \left[-\frac{1}{4}e^{-2x} + \frac{1}{2}e^{-x} - \frac{1}{2}\ln(1+e^{-x}) + C_4 \right] e^{3x}$$

$\therefore$ The general solution is

$$y = y_c + y_p$$

$$= C_1 e^x + C_2 e^{3x} + \left[\frac{1}{2} \ln(1+e^{-x}) + C_3 \right] e^x$$

$$+ \left[-\frac{1}{4}e^{-2x} + \frac{1}{2}e^{-x} - \frac{1}{2}\ln(1+e^{-x}) + C_4 \right] e^{3x}$$

$$= c_1 e^x + c_2 e^{3x} + \frac{1}{2}e^{2x} + \frac{1}{2}e^x \ln(1+e^{-x})$$

$$- \frac{1}{2}e^{3x} \ln(1+e^{-x})$$

where $c_1 = C_1 + C_3 - \frac{1}{4}$ and $c_2 = C_2 + C_4$. (Ans.)

Example-7 Apply the method of variation of Parameter to solve

$$\frac{d^2 y}{dx^2} - y = \frac{2}{1+e^x}$$

Solution The general solution is $y = y_c + y_p = Ae^{-x} + Be^x - e^{-x} \ln(1+e^x) + e^x \ln(1+e^{-x}) - 1$ (Ans.)

Example-8 Solve the following equation by the method of variation of parameters:

$$(D^2 - 3D + 2)y = \cos(e^{-x})$$

Solution Given equation

$$(D^2 - 3D + 2)y = \cos(e^{-x}) \quad \dots (1)$$

The corresponding homogeneous equation is

$$(D^2 - 3D + 2)y = 0 \quad \dots (2)$$

Let $y = e^{mx}$ be a trial solution of (2) then, the auxiliary equation is $m^2 - 3m + 2 = 0$ [$\because e^{mx} \neq 0$]

$$\Rightarrow m = 1, 2$$

$\therefore$ The complementary function is

$$y_c = c_1 e^x + c_2 e^{2x}$$

Let the particular integral of (1) be

$$y_p = Ae^x + Be^{2x} \quad \dots (3)$$

where both A and B are function of x .

Differentiating (3) w. r. to, x we get

$$y_p' = Ae^x + A'e^x + 2Be^{2x} + B'e^{2x} \quad \dots (4)$$

Now we impose the condition

$$A'e^x + B'e^{2x} = 0$$

$$\Rightarrow A' + B'e^x = 0$$

$$\therefore A' = -B'e^x \quad \dots (5)$$

From equation (4), we have,

$$y_p' = Ae^x + 2Be^{2x}$$

$$\therefore y_p'' = Ae^x + A'e^x + 4Be^{2x} + 2B'e^{2x} \quad \dots (6)$$

Now putting the values of y_p, y'_p and y''_p in (1), we get

$$\begin{aligned}
 Ae^x + A'e^x + 4Be^{2x} + 2B'e^{2x} - 3Ae^x - 6Be^{2x} + 2Ae^x + 2Be^{2x} &= \cos(e^{-x}) \\
 \Rightarrow A'e^x + 2B'e^{2x} &= \cos(e^{-x}) \\
 \Rightarrow -B'e^xe^x + 2B'e^{2x} &= \cos(e^{-x}) \quad [\text{using (5)}] \\
 \Rightarrow B'e^{2x} &= \cos(e^{-x}) \\
 \Rightarrow \frac{dB}{dx} &= e^{-2x} \cos(e^{-x}) \quad \dots (7) \\
 \Rightarrow \int dB &= \int e^{-2x} \cos(e^{-x}) dx \quad [\text{Integrating w. r. to } x] \\
 \Rightarrow B &= -\int e^{-x} \cos(e^{-x}) d(e^{-x}) \\
 &= -\int z \cos z \, dz \quad \left| \text{ We put } z = e^{-x} \right. \\
 &= -z \int \cos z \, dz + \int 1 \cdot \sin z \, dz \\
 &= -z \sin z - \cos z, \text{ Omitting integrating constant.} \\
 &= -e^{-x} \sin(e^{-x}) - \cos(e^{-x})
 \end{aligned}$$

Putting the value of B' from (7) into (5), we get

$$\begin{aligned}
 A' &= -e^{-2x} \cos(e^{-x}) e^x \\
 \Rightarrow \frac{dA}{dx} &= -e^{-x} \cos(e^{-x})
 \end{aligned}$$

Integrating this w. r. to x

$$\begin{aligned}
 \Rightarrow \int dA &= -\int e^{-x} \cos(e^{-x}) dx \quad [\text{Integrating both sides}] \\
 &= -\int \cos(e^{-x}) d(e^{-x}) \\
 \therefore A &= \sin(e^{-x}), \text{ Omitting integrating constant.}
 \end{aligned}$$

Now putting the values of A and B , in (3), we get

$$\begin{aligned}
 y_p &= [\sin(e^{-x})]e^x + [-e^x \sin(e^{-x}) - \cos(e^{-x})]e^{2x} \\
 &= -e^{2x} \cos(e^{-x})
 \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p = c_1 e^x + c_2 e^{2x} - e^{2x} \cos(e^{-x})$ (Ans.)

Example-9 Use the method of variation of parameter to solve

$$(D^2 - 2D + 1)y = xe^x \ln x; x > 0$$

Solution The general solution is $y = y_c + y_p = Ae^x + Bxe^x - \frac{5x^3 e^x}{36} + \frac{x^3 e^x \ln x}{6}$, (Ans.)